

A Dheeraj

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Education

RV College of Engineering

B.E. in Mechanical Engineering (CGPA: 8.47)

2022 – 2026

Bangalore, India

Projects

Stock Market Management System | *React.js, Node.js, Express.js, PostgreSQL, Puppeteer*

- Architected a **full stack stock trading application** built with **React, Node.js, Express, and PostgreSQL**, exposing more than **25 REST APIs** supporting authentication, wallet operations, order execution, administrative controls, and AI driven query handling.
- Orchestrated a **low latency transaction processing core** enforcing **100% server side validation** for balance verification, share availability checks, weighted average price computation, and logging across **5 relational tables**.
- Incorporated automation and analytical capabilities including **daily price scraping for greater than 20 companies** via **Puppeteer**, interactive portfolio visualizations (pie/bar/line charts), downloadable performance reports, and an **LLM driven SQL assistant** generating read only statements dynamically.

Real-Time Driver Monitoring System | *MediaPipe, YOLOv8, Streamlit, MongoDB*

- Engineered a **vision based driver safety pipeline** processing **30 video frames per second** to detect drowsiness, yawning, and mobile phone usage through **OpenCV, MediaPipe, and YOLOv8**, triggering live visual and audio alerts.
- Structured a **MongoDB data architecture** with 3 core collections (users, trips, rides) to persist trip level safety events and enable automated generation of **timestamped trip summaries** with alert insights.
- Delivered **role specific analytics interfaces** for drivers and managers, providing fleet wide visibility across 4 critical safety indicators, secure hashed authentication, and driver-manager assignment workflows.

Banking Risk Analysis & Dashboarding | *Python, SQL, Power BI, MySQL*

- Performed exploratory analysis on **3,000+ banking customer records** (25+ attributes) with **Python (pandas, seaborn)**, executing univariate, bivariate, and multivariate evaluations to uncover **0.4–0.6 correlations** between deposits and savings/checking accounts.
- Constructed **5+ derived attributes** (income bands, total loan, total deposit, year joined) and calculated **2 primary KPIs** aggregating **loan exposure** and **total deposits** to facilitate customer risk segmentation.
- Developed a **4-page interactive BI reporting suite** integrated with **MySQL**, comprising **10+ KPI cards, 8+ charts, and 6 slicers** to evaluate loan and deposit behavior across income bands, gender, and relationship categories.

Experience

Amazon ML Summer School | **Selected Participant**

Aug 2025

- Selected among 80,000 applicants for a competitive ML training program where I learnt supervised/unsupervised learning, deep learning architectures (CNNs, sequence models), and deployed ML algorithms using Python frameworks while studying optimization and large scale system design.

Publications & Certifications

Extraction of Biofuel from Algae | *ICDTBESD 2025*

Jul 2025

JPMorgan Chase & Co. — Software Engineering Job Simulation | *SQL, REST APIs*

Sep 2025

Technical Skills

Programming Languages: Java, Typescript, Python, SQL

Web & Backend Technologies: React.js, Node.js, Express.js, REST APIs

Databases: PostgreSQL, MongoDB